



اَوْنُوْرُسِيْتِيْ بِاَتِيْكَوْلُوْ كِيْ بِمَارَا
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FACULTY OF COMPUTING, INFORMATICS AND MATHEMATICS
BACHELOR OF SCIENCE (HONS.) MATHEMATICAL
MODELLING AND ANALYTICS

MAT523 LINEAR ALGEBRA II

GROUP PROJECT CHAPTER 1

TITLE :

***INFANT ABANDONMENT IN MALAYSIA FROM
2008 TO 2017***

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INTRODUCTION

Infant abandonment and infanticide are poorly understood in Malaysia. The information available in the public arena comes predominantly from anecdotal sources. In the last 10 years, there have been almost weekly accounts in local newspapers of abandoned infants.

Total infant abandonment changes and this means that year is independent variable and the total infant abandonment is dependent variable. We are using one mathematical model to find the best fit curve for these data which are quadratic model.

Infant Abandonment Data in Malaysia From 2008 To 2017

Year (X)	Total (Y)
1) 2008	102
2) 2009	79
3) 2010	91
4) 2011	98
5) 2012	89
6) 2013	90
7) 2014	103
8) 2015	111
9) 2016	116
10)2017	120

Table above shows the data about the amount of Infant Abandonment in Malaysia. The data was collect from 2008 until 2017. We took 10 data points from the real details in comparing two variables which are x and y.

X represents the year and Y represents the total amount of infant abandonment in Malaysia from 2008 until 2017.

IMPLEMENTATION

Quadratic Model : $y = a + bx + cx^2$

YEAR (X)	TOTAL (Y)	Y best fit curve	Y error
1	102	94.3909	7.6091
2	79	91.5545	-12.5545
3	91	90.2409	0.7591
4	98	90.4500	7.5500
5	89	92.1818	-3.1818
6	90	95.4364	-5.4364
7	103	100.2136	2.7864
8	111	106.5136	4.4864
9	116	114.3364	1.6636
10	120	123.6818	-3.6818

The data generates a system of quadratic equations with the matrix representation as below:

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \\ 1 & 4 & 16 \\ 1 & 5 & 25 \\ 1 & 6 & 36 \\ 1 & 7 & 49 \\ 1 & 8 & 64 \\ 1 & 9 & 81 \\ 1 & 10 & 100 \end{bmatrix} \mathbf{v} = \begin{bmatrix} 102 \\ 79 \\ 91 \\ 98 \\ 89 \\ 90 \\ 103 \\ 111 \\ 116 \\ 120 \end{bmatrix}$$

Its normal equation is

$$\begin{bmatrix} 10 & 55 & 385 \\ 55 & 385 & 3025 \\ 385 & 3025 & 25333 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 999 \\ 5763 \\ 41817 \end{bmatrix}$$

The least square solution is

$$v = \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} \frac{83}{60} & -\frac{21}{40} & \frac{1}{24} \\ -\frac{21}{40} & \frac{637}{2640} & -\frac{1}{48} \\ \frac{1}{24} & -\frac{1}{48} & \frac{1}{528} \end{bmatrix} \begin{bmatrix} 999 \\ 5763 \\ 41817 \end{bmatrix} = \begin{bmatrix} \frac{395}{4} \\ -\frac{2253}{440} \\ \frac{67}{88} \end{bmatrix} = \begin{bmatrix} 98.75 \\ -5.120454545 \\ 0.7613636364 \end{bmatrix}$$

The best fit quadratic curve is **$y = 98.75 - 5.12x + 0.76x^2$**

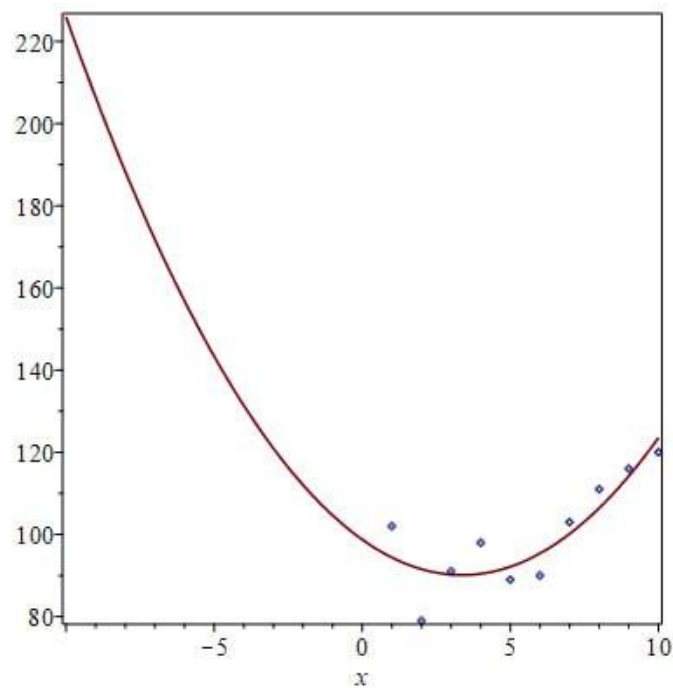
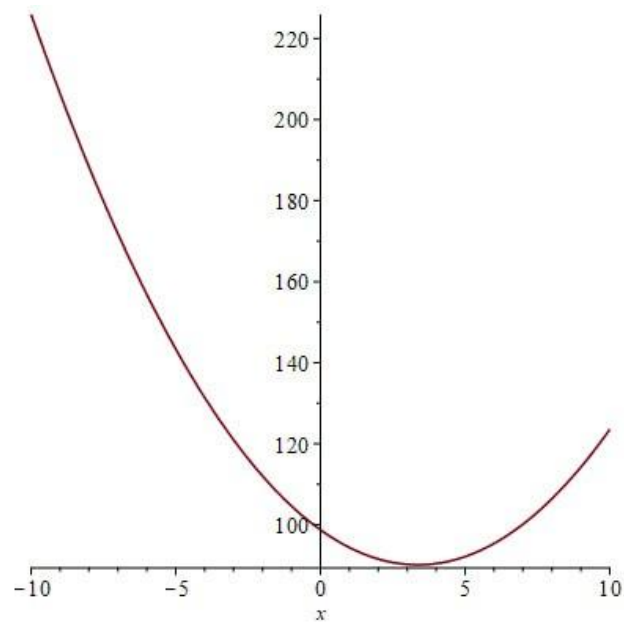
The error vector is

$$e = y - Mv = \begin{bmatrix} 102 \\ 79 \\ 91 \\ 98 \\ 89 \\ 90 \\ 103 \\ 111 \\ 116 \\ 120 \end{bmatrix} - \begin{bmatrix} \frac{10383}{110} \\ \frac{10071}{110} \\ \frac{19853}{220} \\ \frac{1809}{20} \\ \frac{1014}{11} \\ \frac{5249}{55} \\ \frac{22047}{220} \\ \frac{23433}{220} \\ \frac{12577}{110} \\ \frac{2721}{22} \end{bmatrix} = \begin{bmatrix} \frac{837}{110} \\ -\frac{1381}{110} \\ \frac{167}{220} \\ \frac{151}{20} \\ -\frac{35}{11} \\ -\frac{299}{55} \\ \frac{613}{220} \\ \frac{987}{220} \\ \frac{183}{110} \\ -\frac{81}{22} \end{bmatrix}$$

$$\|e\| = \frac{1381}{110}$$

$$= 12.55454545$$

GRAPHS



This is a quadratic graph of $y = 98.75 - 5.12x + 0.76x^2$ which is a polynomial degree of two. From the graph, we can conclude that the higher the year, the infant abandonment data in Malaysia will be increased.

ANALYSIS AND CONCLUSION

Based on model above, the following error magnitudes in the approximation are obtained.

The best fit curve for this case which will give the best approximation is

$y = 98.75 - 5.12x + 0.76x^2$ with the error magnitude of 12.5545.

This is because it has the minimum error in terms of its magnitude/norm obtained from calculation above.

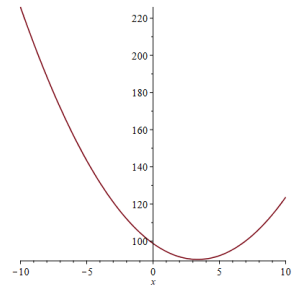
APPENDIX

> restart	
> with(LinearAlgebra):	
> interface(rtobsize = infinity)	
> M := Matrix([[1, 1, 1], [1, 2, 4], [1, 3, 9], [1, 4, 16], [1, 5, 25], [1, 6, 36], [1, 7, 49], [1, 8, 64], [1, 9, 81], [1, 10, 100]])	10
	$M := \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \\ 1 & 4 & 16 \\ 1 & 5 & 25 \\ 1 & 6 & 36 \\ 1 & 7 & 49 \\ 1 & 8 & 64 \\ 1 & 9 & 81 \\ 1 & 10 & 100 \end{bmatrix}$
> y := Matrix([[102], [79], [91], [98], [89], [90], [103], [111], [116], [120]])	
	$y := \begin{bmatrix} 102 \\ 79 \\ 91 \\ 98 \\ 89 \\ 90 \\ 103 \\ 111 \\ 116 \\ 120 \end{bmatrix}$
> MT := Transpose(M)	
	$MT := \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 \\ 1 & 4 & 9 & 16 & 25 & 36 & 49 & 64 & 81 & 100 \end{bmatrix}$
> (MT.M)	
	$\begin{bmatrix} 10 & 55 & 385 \\ 55 & 385 & 3025 \\ 385 & 3025 & 25333 \end{bmatrix}$
> (MT.y)	
	$\begin{bmatrix} 999 \\ 5763 \\ 41817 \end{bmatrix}$
> ((MT.M)) ⁻¹	
	$\begin{bmatrix} \frac{83}{60} & -\frac{21}{40} & \frac{1}{24} \\ -\frac{21}{40} & \frac{637}{2640} & -\frac{1}{48} \\ \frac{1}{24} & -\frac{1}{48} & \frac{1}{528} \end{bmatrix}$
> v := ((MT.M)) ⁻¹ .(MT.y)	
	$v := \begin{bmatrix} \frac{395}{4} \\ -\frac{2253}{440} \\ \frac{67}{88} \end{bmatrix}$
> (M.v)	
	$\begin{bmatrix} \frac{10383}{110} \\ \frac{10071}{110} \\ \frac{19853}{220} \\ \frac{1809}{20} \\ \frac{1014}{11} \\ \frac{5249}{55} \\ \frac{22047}{220} \\ \frac{23433}{220} \\ \frac{12577}{110} \\ \frac{2721}{110} \end{bmatrix}$


```
R> with(Statistics) :
R> X := [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
X := [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
R> Y := [102, 79, 91, 98, 89, 90, 103, 111, 116, 120]
Y := [102, 79, 91, 98, 89, 90, 103, 111, 116, 120]
R> b := ScatterPlot(X, Y)
```



```
• a := plot( $\frac{395}{4} - \frac{2253}{440}x + \frac{67}{88}x^2$ )
```



```
• plots[display](a, b)
```

